

EPIGO: Entropy and Spectral Structure of Epigenomic Signal Tracks

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Abstract

Epigenomic assays generate quantitative, genome-indexed signal tracks that summarize chromatin state and regulatory activity across genomic coordinates. This document introduces epigenetic signal tracks, presents the EPIGO project as a mathematically grounded framework for characterizing such tracks, and formalizes the core computations—entropy and one-dimensional spectral structure—that underlie EPIGO’s feature construction and comparison pipeline.

1 Introduction to Epigenetics

Epigenetics concerns molecular mechanisms that modulate genome function without altering the underlying DNA sequence. In eukaryotes, these mechanisms operate through chromatin: DNA packaged with histone proteins into nucleosomes and higher-order structures. The functional state of chromatin is influenced by (i) *DNA methylation*, typically the covalent addition of methyl groups to cytosines in context-dependent patterns; (ii) *histone modifications*, chemical marks such as acetylation or methylation on histone tails that recruit or repel regulatory machinery; and (iii) broader *chromatin marks* and associated protein complexes that collectively shape accessibility, compaction, and long-range organization.

A defining property of modern epigenomics is that these mechanisms are observed through high-throughput assays that produce *quantitative genomic tracks*¹: real-valued signals indexed by genomic coordinate. Examples include ChIP-seq enrichment tracks for histone marks, accessibility tracks from DNase-seq or ATAC-seq, and methylation-derived measures from bisulfite sequencing. Conceptually, such data are not merely collections of discrete loci; rather, they are spatially structured signals distributed along chromosomes. Consequently, an epigenomic dataset can be viewed as a family of quantitative functions over the genome, with meaningful variation in both amplitude and spatial organization across scales.

2 Overview of the EPIGO Project

EPIGO (Epigenetic Information Geometry Observatory; alternatively framed as *Epigenetics, built like LEGO*) is a mathematical and computational framework for analyzing and visualizing structured biological signals—in particular, epigenomic tracks—using information-theoretic and spectral methods, with an emphasis on interpretability, composition, and explicit assumptions (https://github.com/cluehning/march_EPIGO).

¹EPIGO adopts a signal-centric view of epigenomic data, treating tracks as spatially extended objects rather than collections of independent genomic sites. This perspective is consistent with large-scale epigenomic mapping efforts, where chromatin marks are measured continuously across the genome and interpreted in terms of regional organization rather than single positions [4, 6].

The core idea is to represent an epigenomic track as a discrete signal obtained by binning genomic coordinates, to convert the resulting intensities into a probability distribution, and to compute complementary descriptors:

1. *Information-theoretic structure*: Shannon entropy summarizes how evenly signal mass is distributed across bins ², providing a scale-aware notion of diffuseness versus concentration.
2. *Spectral structure*: a one-dimensional discrete Fourier analysis characterizes spatial organization, separating contributions from different genomic length scales and revealing periodicity or roughness.

These computations produce feature representations that can be compared quantitatively and visualized consistently across tracks ³.

2.1 Pipeline modules

EPIGO is organized around modular steps that map directly to numerical operations:

- **Signal model and binning**: convert interval-based signals (e.g. bedGraph derived from bigWig) into uniformly binned vectors representing average intensity per genomic bin.
- **Shannon entropy computation**: normalize binned intensities into a probability vector (with ε -smoothing) and compute entropy and normalized entropy.
- **1D spectral analysis**: center the signal by removing the mean, compute the discrete Fourier transform (DFT), and derive a power spectrum; optionally apply smoothing (Welch-style averaging) to reduce variance.
- **Feature construction and comparison**: assemble entropy and spectral summaries into a feature vector and compare tracks via distances (e.g. Euclidean) or, when using probability representations, divergences such as Kullback–Leibler.

3 Mathematical Foundations (based on the provided notes)

This section formalizes the objects and computations used by EPIGO, following implementation-agnostic definitions aligned with the pipeline’s numerical operations.

3.1 Genomic signal tracks as functions

An epigenomic signal track is modeled as a genome-indexed scalar field

$$x : \{(c, p)\} \rightarrow \mathbb{R}_{\geq 0}, \quad (1)$$

where c denotes a chromosome identifier and p a genomic coordinate (in base pairs). The quantity $x(c, p)$ represents signal intensity at position p on chromosome c .

In interval-based formats such as bedGraph, the signal is represented as a piecewise-constant function along each chromosome:

$$x(p) = v_i \quad \text{for } p \in [s_i, e_i), \quad (2)$$

where each row (c, s_i, e_i, v_i) encodes a constant value v_i on the half-open interval $[s_i, e_i)$.

²In EPIGO, entropy is not interpreted as biological “complexity” in a strong sense. Rather, it is a descriptive statistic of distributional uniformity. High entropy can arise from noise as well as from broadly distributed regulatory activity, and must be interpreted in context.

³Internally, EPIGO treats each epigenomic track as a modular object with well-defined mathematical properties. The informal “LEGO” metaphor reflects this modularity: tracks can be measured, compared, and combined without collapsing them into a single predictive task.

3.2 Binning and discrete representation

For numerical analysis, EPIGO converts the interval representation into a uniformly sampled discrete signal by binning genomic positions into fixed-width bins of size B base pairs⁴. For bin index k , the binned signal value is defined by the bin average

$$x_k = \frac{1}{B} \int_{kB}^{(k+1)B} x(p) dp \approx \frac{1}{B} \sum_{p=kB}^{(k+1)B-1} x(p). \quad (3)$$

Over a region, chromosome, or concatenated genome, this yields a vector

$$x = (x_0, \dots, x_{N-1}) \in \mathbb{R}_{\geq 0}^N,$$

which serves as the discrete representation of the track.

3.3 Probability normalization with ε -smoothing

Entropy is defined on probability distributions. EPIGO therefore maps intensities to a probability vector by normalized mass assignment with additive smoothing:

$$p_k = \frac{x_k + \varepsilon}{\sum_{j=0}^{N-1} (x_j + \varepsilon)}, \quad \varepsilon > 0. \quad (4)$$

Here p_k represents the fraction of total (smoothed) signal mass in bin k . The constant ε ensures numerical stability by preventing $p_k = 0$, avoiding undefined logarithms.

3.4 Shannon entropy and normalized entropy

Given $p = (p_0, \dots, p_{N-1})$ with $p_k \geq 0$ and $\sum_k p_k = 1$, Shannon entropy is

$$H(p) = - \sum_{k=0}^{N-1} p_k \log p_k. \quad (5)$$

In this setting, $H(p)$ quantifies how evenly the signal mass is distributed across genomic bins: higher entropy indicates a more diffuse signal, while lower entropy indicates concentration in fewer bins.

To compare regions with different N , a normalized entropy is often used:

$$H_{\text{norm}}(p) = \frac{H(p)}{\log N}, \quad (6)$$

which bounds the value to $[0, 1]$ under the usual conventions.

3.5 Centered signal, DFT, and power spectrum

To characterize spatial structure, EPIGO analyzes the discrete signal in the frequency domain. First remove the mean (detrending/centering):

$$\tilde{x}_k = x_k - \bar{x}, \quad \bar{x} = \frac{1}{N} \sum_{k=0}^{N-1} x_k. \quad (7)$$

⁴Binning is not merely a technical convenience but an explicit modeling assumption: it defines the spatial resolution at which structure is considered meaningful. All entropy and spectral quantities in EPIGO are therefore scale-dependent, and comparisons are only interpretable when binning choices are held fixed or analyzed systematically.

This removes the zero-frequency (DC) component so that spectral content reflects spatial variation rather than absolute level ^{5 6}.

The discrete Fourier transform (DFT) of the centered signal is

$$X_m = \sum_{k=0}^{N-1} \tilde{x}_k e^{-i2\pi mk/N}, \quad m = 0, \dots, N-1, \quad (8)$$

and the (raw) power spectrum (periodogram) is defined by

$$S_m = \frac{1}{N} |X_m|^2. \quad (9)$$

Since $\tilde{x}_k$ is real-valued, the spectrum is symmetric; analyses commonly retain a one-sided spectrum.

If each bin spans B base pairs, the spatial frequency corresponding to index m is

$$f_m = \frac{m}{NB} \quad (\text{cycles per base}), \quad (10)$$

with characteristic wavelength $\lambda_m \approx 1/f_m$ (interpreted at nonzero frequencies).

3.6 Optional smoothing (Welch-style averaging)

Raw periodograms exhibit high variance. EPIGO may reduce variance by averaging neighboring frequency bins with weights $(w_r)_{r=-R}^R$ satisfying $\sum_{r=-R}^R w_r = 1$:

$$\hat{S}_m = \sum_{r=-R}^R w_r S_{m+r}. \quad (11)$$

This smoothing produces a more stable estimate of spectral structure at the cost of frequency resolution.

4 What the Project Does in Practice

In typical usage, EPIGO ingests epigenomic tracks stored as bigWig or bedGraph ⁷. Because bigWig is a compressed binary format optimized for efficient access, a common workflow is to convert bigWig to bedGraph using established utilities, then process the resulting interval-based text representation. EPIGO then:

1. **Loads and bins the track:** interval values are aggregated into fixed-width bins of size B , producing a discrete vector x of average intensities.
2. **Computes entropy features:** x is mapped to a probability vector p via ε -smoothed normalization, and $H(p)$ (and optionally $H_{\text{norm}}(p)$) is computed. Entropy provides a single-number summary of signal diffuseness ⁸: broad, weakly concentrated tracks yield higher entropy, whereas sharp enrichment yields lower entropy.

⁵Centering eliminates the zero-frequency (DC) component so that spectral power reflects spatial variation rather than absolute signal level. Without this step, differences in overall enrichment would dominate low-frequency power and obscure genuine organizational structure, a standard consideration in spectral estimation [8].

⁶Peak calling methods such as MACS introduce hard thresholds that collapse continuous signals into binary or sparse representations [9]. EPIGO avoids this step in order to preserve weak, distributed, or multiscale structure that may be biologically meaningful even when it does not form sharp peaks.

⁷The bigWig format is a compressed, indexed representation designed for large genomic signal datasets [7]. For numerical analysis, tracks are commonly converted to bedGraph using standard UCSC utilities distributed via Bioconda [2].

⁸In EPIGO, Shannon entropy is used as a descriptive statistic of distributional uniformity rather than as a measure of biological complexity in a strong sense. While entropy has a precise information-theoretic definition [5], its interpretation here follows standard treatments emphasizing uncertainty and mass dispersion in discrete distributions [1, 3].

3. **Computes spectral features:** the centered signal $\tilde{x}$ is transformed by the DFT and summarized through the power spectrum (optionally smoothed). The spectrum captures spatial structure⁹, including characteristic scales, periodicity, and roughness. Tracks with comparable entropy can nonetheless differ strongly in their spectral organization.
4. **Constructs feature vectors and compares tracks:** EPIGO combines information-theoretic and spectral descriptors¹⁰ into a feature map

$$\phi(x) = (H(p(x)), \hat{S}(x), \text{summary statistics}), \quad (12)$$

and compares tracks using distances such as

$$d(\phi_1, \phi_2) = \|\phi_1 - \phi_2\|_2, \quad (13)$$

or, when interpreting tracks as distributions, divergences such as the Kullback–Leibler divergence

$$D_{\text{KL}}(p\|q) = \sum_k p_k \log \frac{p_k}{q_k}. \quad (14)$$

Together, these steps provide a reproducible workflow for characterizing and contrasting epigenomic tracks as structured objects rather than as collections of isolated test statistics.

5 Relation to the Classical Methylation Analysis

Classical differential methylation analysis typically focuses on locus-by-locus inference: identifying individual CpG sites or genomic regions whose methylation levels differ significantly between conditions, often via statistical tests and multiple-testing correction. The primary output is a set of sites/regions with estimated effect sizes and significance measures, supporting localized biological interpretation and downstream enrichment analyses.

EPIGO differs in emphasis and representation. Rather than treating the track as a set of independent loci, EPIGO treats methylation- or chromatin-derived measurements as a *spatial signal* and focuses on *signal structure*: how mass is distributed (entropy) and how variation is organized across genomic scales (spectrum). This is not a substitute for differential methylation testing; instead, it complements locus-level approaches¹¹ by providing global and mesoscopic descriptors that can reveal differences in organization even when no single locus dominates the effect.

Despite these differences, both approaches share a common goal: characterizing epigenomic variation across samples, conditions, or cell types. Differential methylation targets localized change detection; EPIGO targets quantitative, scale-aware summaries of track-wide structure.

6 Boundaries, Assumptions, and Sources of Error

As with any quantitative pipeline, EPIGO’s outputs depend on modeling choices and data quality. The following issues can lead to mistakes, instability, or misinterpretation.

⁹In practice, spectral features reflect properties such as smoothness, roughness, and characteristic length scales of variation. They should not be interpreted as literal periodic biological processes unless supported by independent evidence.

¹⁰Entropy and spectral structure are intentionally paired: entropy summarizes how signal mass is distributed, while the spectrum captures how that mass is organized across spatial scales. Either alone is incomplete; together they distinguish diffuse versus concentrated signals and smooth versus oscillatory structure.

¹¹EPIGO does not test hypotheses about individual loci and does not assign statistical significance to specific genomic positions. Its role is descriptive and comparative, providing global and mesoscopic summaries that can motivate or contextualize downstream locus-level analyses.

6.1 Choice of bin size B

The bin width B controls resolution and variance. Larger bins emphasize coarse-scale structure and reduce noise, but can blur narrow peaks and shift spectral content toward lower frequencies by aggregation. Smaller bins preserve fine structure but increase variance in both entropy (through sparse mass distributions) and spectral estimates (through noisier $\tilde{x}$ and higher-frequency components). Consequently, meaningful comparisons require consistent B (or explicit multiscale analysis).

6.2 Zero-handling and ε -smoothing

Entropy involves $\log p_k$ and is therefore sensitive to zero probabilities. EPIGO uses additive ε -smoothing in the normalization

$$p_k \propto x_k + \varepsilon,$$

which stabilizes computation but introduces a bias: very small ε approximates the raw distribution yet may remain numerically fragile, while larger ε pushes p toward uniformity, artificially increasing entropy and attenuating differences among tracks. Moreover, entropy depends on normalization: global rescaling of x cancels under exact normalization, but the presence of ε makes entropy sensitive to the absolute scale of x when typical x_k values are comparable to ε .

6.3 Finite-window effects

When analysis is restricted to genomic windows, N is finite and boundaries matter. Sharp truncation can introduce spectral leakage, redistributing power across frequencies and creating artifacts that resemble structure. Comparisons across windows require consistent window definitions, and aggregating window-level summaries should account for heterogeneity across genomic contexts.

6.4 High-variance raw periodograms

The raw periodogram S_m is a noisy estimator of power. Without smoothing, two realizations of similar underlying structure can yield substantially different S_m due to sampling variability and finite-length effects. Welch-style averaging reduces variance but also reduces frequency resolution; the choice of smoothing radius and weights thus trades stability for detail.

6.5 Detrending and window length

Spectral estimates depend on detrending choices. Mean removal eliminates the DC component but does not address other trends (e.g. slow drifts across a region) that can inflate low-frequency power. Additionally, window length (equivalently N) governs frequency resolution: short windows coarsen spectral bins and can mask narrowband structure, while long windows increase resolution but may mix heterogeneous genomic regimes.

6.6 Biological caveats: noise and coverage

Epigenomic tracks can be affected by assay noise, sequencing depth, mappability, and preprocessing choices (alignment, normalization, peak calling ¹², background correction). For example, sparse coverage or variable depth can induce apparent concentration or roughness unrelated to biology. Similarly, different track types (p-value tracks, fold-enrichment, counts) embed distinct transformations that change amplitude distributions and, consequently, derived entropy

¹²Peak calling imposes hard thresholds and discards subthreshold structure. EPIGO deliberately avoids this step in order to preserve weak, distributed, or multiscale patterns that may be biologically meaningful even when they do not form sharp peaks.

and spectral features. Interpretations should therefore be conditioned on the provenance and preprocessing of the input tracks.

6.7 Scope limitations

EPIGO is descriptive: it summarizes and compares track structure. It does not infer causality, mechanism, or biological function by itself ¹³, nor does it replace experimental validation. Any biological interpretation must integrate context (cell type, mark identity, experimental design) and should avoid overextending mathematical descriptors beyond what the data support.

7 Conclusion

EPIGO provides a reproducible, mathematically grounded framework for characterizing epigenomic tracks as structured, quantitative signals. By combining information-theoretic measures (entropy) with frequency-domain descriptors (power spectra), EPIGO enables comparisons that are sensitive to both distributional diffuseness and spatial organization across genomic scales. This perspective complements locus-centric analyses by elevating track-level structure to a first-class object of study, supporting interpretability, modularity, and systematic exploration of epigenomic variation.

¹³All quantities produced by EPIGO are descriptive summaries of observed signal structure. While these summaries can motivate hypotheses or guide comparisons, causal or mechanistic claims require external biological knowledge and independent experimental validation.

8 Phase 2: Extension to Pairwise and Localized Signal Analysis

Phase 2 extends the original EPIGO framework from single-track summaries to a broader analysis of *relationships* and *local structure* within and between epigenomic signals. While Phase 1 focused on describing individual tracks via entropy and spectral structure, Phase 2 asks additional questions:

- How similar are two tracks in their distribution of signal?
- At what spatial scales do two tracks share structure?
- Where along the genome do these relationships appear or disappear?

These extensions preserve the original signal-centric perspective but introduce a more relational and position-dependent view of epigenomic data.

8.1 Conceptual Overview

The key shift in Phase 2 is the interpretation of tracks as *empirical probability distributions*, allowing the use of classical information-theoretic comparisons. Once signals are discretized into bins and normalized, they can be compared not only individually, but as *pairs of distributions*.

In parallel, spectral analysis is extended from describing a single signal to describing *shared structure across signals*, and from global summaries to *localized behavior*.

Conceptually:

- Phase 1: “What does this track look like?”
- Phase 2: “How do these tracks relate, and where do those relationships occur?”

8.2 Information-Theoretic Comparisons

Phase 2 introduces pairwise quantities such as mutual information and divergence measures (Section 6.1). These operate on discretized distributions derived from the underlying signals.¹⁴

Mutual information provides a measure of how much knowing one signal reduces uncertainty about another. In this setting, it should be interpreted as a *measure of shared structure*, rather than a statement about direct biological interaction.

Similarly, Jensen–Shannon divergence provides a symmetric and finite comparison between distributions, allowing a notion of “distance” between tracks that is more stable than raw Kullback–Leibler divergence.¹⁵

A key point is that all of these quantities depend on discretization choices (bin size, binning strategy) and therefore represent *empirical approximations* rather than intrinsic properties of the underlying continuous signal.

¹⁴As in Phase 1, distributions are obtained via binning and normalization with ε -smoothing to ensure numerical stability.

¹⁵Because JSD averages distributions before computing divergence, it avoids the instability associated with zero-probability bins.

8.3 Spectral Relationships Between Tracks

In addition to distributional comparisons, Phase 2 extends spectral analysis to pairs of signals through coherence (Section 6.2). This allows identification of shared frequency or scale structure.

“Do two tracks vary together at the same spatial scales?”

Coherence is bounded between 0 and 1 and is computed from cross-spectral and auto-spectral densities. It should be interpreted descriptively as evidence of shared organization across scales, not as evidence of causality.¹⁶

8.4 Localized (Windowed) Structure

A major addition in Phase 2 is the ability to compute quantities locally along the genome (Section 6.3).

Instead of computing entropy or mutual information once over an entire signal, the signal is divided into overlapping or non-overlapping windows. Each window produces a scalar summary, such as:

- entropy,
- mutual information,
- spectral energy,
- coherence summary.

This produces position-dependent profiles of the form:

$$\{(c_k, v_k)\}_k,$$

where c_k is the genomic position of the window and v_k is the computed metric.

This extension allows EPIGO to move from global summaries to *regional structure*, making it possible to identify where signals are concentrated, correlated, or structurally similar along the genome.¹⁷

8.5 Localized Frequency Analysis (STFT)

To handle signals whose structure changes across position, Phase 2 introduces the short-time Fourier transform (STFT) (Section 6.4).

Rather than treating the signal as globally stationary, STFT computes frequency content within a moving window:

$$X(t, f) = \sum_n x[n] w[n - t] e^{-i2\pi f n}.$$

This yields a joint representation in position and frequency, capturing how dominant scales evolve along the signal.¹⁸

¹⁶As in Phase 1, spectral features describe spatial organization and should not be interpreted as direct biological mechanisms without external validation.

¹⁷Windowing introduces a new scale parameter (window size), which must be chosen consistently for meaningful comparison.

¹⁸STFT trades frequency resolution for spatial localization, making it appropriate for signals with non-uniform structure.

8.6 Limitations of the Phase 2 Extension

While Phase 2 significantly extends the expressive power of EPIGO, it also introduces additional assumptions and potential sources of error.

8.6.1 Discretization dependence

All information-theoretic quantities depend on binning strategy (number of bins, bin edges, quantile vs uniform binning). Different choices can yield different numerical values, especially for mutual information and divergence measures.

8.6.2 Finite data and sampling effects

All estimates are empirical. Small sample sizes or highly uneven distributions can lead to unstable or biased estimates, even with ε -smoothing.

8.6.3 Smoothing bias

The use of ε -smoothing ensures numerical stability but biases distributions toward uniformity. This can increase entropy and reduce apparent differences between tracks.

8.6.4 Windowing tradeoffs

Localized analysis introduces a tradeoff:

- small windows $\rightarrow$ high spatial resolution, high variance
- large windows $\rightarrow$ lower variance, reduced localization

Results depend strongly on the chosen window size.

8.6.5 Spectral estimation variance

Spectral and cross-spectral estimates can be noisy for finite-length signals. Smoothing improves stability but reduces resolution in frequency space.

8.6.6 Interpretation of dependence

Measures such as mutual information and coherence capture *statistical dependence*, not causation. Shared structure may arise from indirect effects, confounding variables, or shared preprocessing artifacts.¹⁹

8.6.7 Non-stationarity assumptions

Although STFT addresses non-stationarity, it still relies on local stationarity within each window. Rapidly changing signals may not be well captured at any single window scale.

¹⁹EPIGO remains a descriptive framework; biological interpretation requires external validation and domain knowledge.

8.7 Summary

Phase 2 extends EPIGO from describing individual tracks to analyzing:

- relationships between tracks,
- spatial variation in structure,
- scale-dependent interactions.

These additions preserve the original framework but expand it into a more general tool for exploring structured biological signals across multiple levels: distributional, spectral, and spatial.

The core principle remains unchanged:

EPIGO provides quantitative, interpretable summaries of signal structure — not causal explanations.

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